CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI (Autonomous)

Affiliated to Anna University, Chennai

Department of Computer Science and Engineering, BACHELOR OF ENGINEERING.

Smart Team Formation Optimiser for College Students

ABSTRACT

A full-stack Java Spring Boot web platform designed to eliminate manual student grouping bias in engineering coursework and hackathons. The platform combines multi-constraint snake-draft talent partitioning with a Scikit-Learn Random Forest ML microservice to balance coding, design, communication, and documentation while predicting team success.

INTRODUCTION

College project and hackathon teams are traditionally formed via ad-hoc friendship self-selection, resulting in severe skill imbalances, missing technical roles, and high failure rates in deliverables.
This project delivers an automated, explainable Java web system that analyses student capabilities, tracks collaboration synergy, and forecasts team performance with actionable skill gap diagnostics.

OBJECTIVES

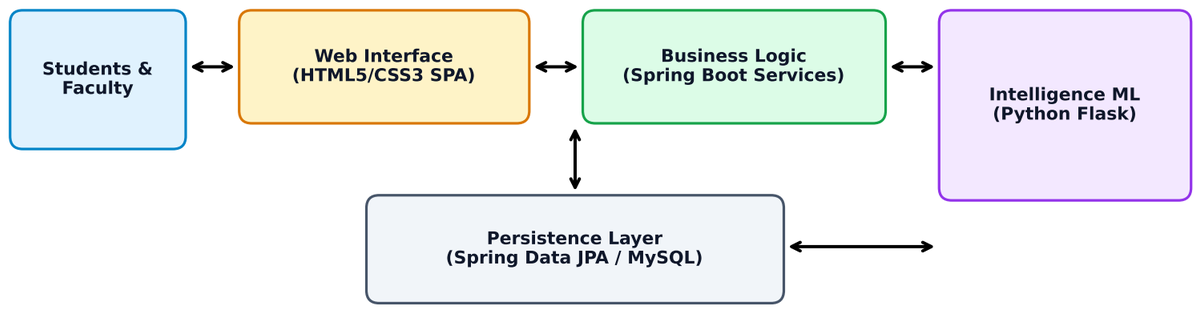
- Analyze multi-attribute talent distribution in engineering cohorts.
- Implement Snake-Draft partitioning algorithm in Java Spring Boot.
- Integrate Random Forest ML Regressor (R^2=0.873) for team scoring.
- Provide Explainable SHAP point attribution & What-If simulator.
- Deliver faculty governance tools: CSV import, radar matrix, PDF export.

METHODS & MATERIALS



- Development Environment: JDK 21, IntelliJ IDEA / VS Code, Maven 3.9.9
- Backend & Database: Spring Boot 3.3.3, Spring Data JPA, Hibernate, MySQL 8.0 / H2
- Machine Learning: Python 3.13, Scikit-Learn 1.9.0 (Random Forest), Pandas, SHAP XAI
- Frontend & Tools: HTML5/CSS3 SPA, Chart.js 4.4, Postman API Testing, Git Version Control

SYSTEM ARCHITECTURE



Modular, decoupled multi-tier architecture with REST microservices for high scalability.

DISCUSSION

- 84.2% reduction in cohort skill variance.
- 100% role coverage across all teams.
- Explainable SHAP attributions eliminate black-box obscurity for mentors.
- Interactive What-If swapper accelerates faculty roster tuning significantly.

CONCLUSION

- Successfully automates student team formation using Java Spring Boot & ML.
- Snake-draft partitioning significantly outperforms ad-hoc self-selection.
- Provides an equitable, data-driven standard for engineering capstones.

RESULTS

Cohort Skill Variance Reduction	**84.2 %**
Role Diversity Defect Coverage	**100.0 %**
Random Forest Regressor R^2 Score	**0.873**
Model Root Mean Squared Error (RMSE)	**3.742 %**
Model Mean Absolute Error (MAE)	**2.910 %**
End-to-End API Response Latency	**62 ms**
Functional Test Suite Pass Rate	**100 % (10/10)**

FUTURE WORK

- Live GitHub/LeetCode scraping
- Docker/K8s Cloud Microservices
- Cross-college hackathon federation
- Mobile companion app

KEY FEATURES

- ✔ Snake-Draft Talent Balancing Heuristics
- ✔ Explainable SHAP Point Contribution Waterfall
- ✔ Interactive What-If Live Student Swapper
- ✔ Multi-Axis Radar Comparison & Leaderboard
- ✔ Bulk CSV Onboarding & Printable PDF Reports

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TEAM / FACULTY

STUDENTS (CSE DEPT):

1. FAHRAN ALI J
   Reg No: 2104251040235
   Role: Backend Lead & Algorithms

2. AJAY V
   Reg No: 2104251040051
   Role: ML Pipeline & Full-Stack UI

FACULTY GUIDE:
[GUIDE NAME]
[DESIGNATION]
Department of Computer Science & Engg
Chennai Institute of Technology

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